CIVL1113 Engineering
Mechanics and Materials
May 2025 Examination
Comments on student performance

Mechanics part

General: Mechanics part

- The performance of students varies greatly. In general, many students were able to solve the problem and showed basic understanding on the course contents in mechanic part.
- A problem of some students is the lack of understanding of internal forces in a structure, which is related to inability to construct bending moment diagram (BMD) and shear force diagram (SFD) correctly. Looking at BMDs and SFDs for typical cases will help.
- Some students did not have the ability to spot problems from unusual results, e.g., convert between different unit systems, method of section, and follow up with further checking.
- Some students had difficulty in remembering the basic equations for the calculation, which had been emphasized many times in courses. It is strongly recommended that all student attend all the course lectures to understand the basic knowledge, otherwise it is very hard to understand the complicated mechanics course concepts covered in this course. Some basic equations are noted as very important equations and marked in red in lecture notes.
- Looking at the previous semester's tutorial videos recorded may help, especially for buckling.

Question 1

- The performance in this question was fine but varied.
- In Question 1(a), the performance was generally fine but there are still some students who did not work out the results clearly. The governing equation of torsion needs to be remembered.
- Question 1(b), students should be able to understand the force transfer in this structure and use method of section to work out the SFD and BMD. Not all students follow the steps to obtain the results.
- Some students presented BMD and SFD of wrong shapes, suggesting inadequate understanding of the concepts. It is recommended to practice more on relatively simple statically determinate structures.
- Some students showed inadequate understanding of the sign conventions in the construction of BMD and SFD as guided by the exam questions. For connection between one single column and one beam, the bending moment should be equal.

Question 2

- The overall performance in this question was fine. The performance varied.
- Some students had some difficulty in converting the unit systems in Q2(a).
- Some students used the shear force equation of rectangular beam, which is not correct.
- Some students did not check the safety at the connection between web and flange.
- For the stress transformation problem, either the stress transformation equation or the Mohr's circle can be used. However, the stress transformation equation will be much easier.
- Some students did not use the strength criteria to check the safety as requested by the exam paper.

Question 3

- The overall performance in this question was moderate. The effective length of this problem

is 2 times the original length. Actually, this is the same as a cantilever column, the only difference is that the top horizontal displacement is fixed while the bottom horizontal displacement is free. Some students remembered Euler's equation incorrectly. Some students did not work out the effective length correctly. Out-of-plane buckling does not need to be considered for this problem. Some students did not remember the polar moment inertia of the cross-section. Looking at some other questions or textbooks may help improve understanding.